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Progressive Mask Distillation for Self-supervised Video Representation

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Abstract

Masked visual modeling is a self-supervised learning task that does not use visual annotations. It aims to learn discriminative representations via a mask-reconstruction task. A single mask ratio in reconstruction may fail to capture complex semantics, which motivates dynamic masking strategies. In this work, we propose Progressive Mask Distillation (PMD), which utilizes dynamic mask ratios to facilitate progressive semantic learning from easy to hard. PMD integrates three key components: a progressive student distiller, a difficulty-aware region enhancer, and a cross-layer feature aligner. First*, to capture dynamic visual semantics, we design a progressive student distiller that trains multiple student models with progressively increasing mask ratios. The early-phase student (with a low mask ratio) learns easy, low-level semantics from more visible tokens. This learned knowledge then guides the next-phase student (with a higher mask ratio) to capture hard, high-level semantics from fewer visible tokens. This progressive distillation mechanism enhances detail reconstruction at a high mask ratio.* Second*, to alleviate insufficient learning of semantic regions, we design a difficulty-aware region enhancer. It uses the region reconstruction loss to learn their weights, prioritizing accurate learning of regions with large reconstruction losses.* Third*, to further bridge the semantic gap across network layers, we design cross-layer feature alignment. This module aligns features across shallow, middle, and deep encoder layers, ensuring that shallow-layer features incorporate semantic information from deeper layers. Extensive experiments demonstrate that our method achieves state-of-the-art performance on the SSv2, K400, UCF-101, and HMDB-51 datasets.*

1. Introduction

Masked visual modeling is a self-supervised learning task that does not rely on visual annotations. It learns visual representation by reconstructing targets from masked inputs, and provides effective features for downstream tasks (e.g., action recognition [30]). Existing methods adopt different masking strategies to model spatiotemporal relations for self-supervised representation learning [8, 30, 32]. Some methods enhance masking strategies with motion trajectories [29] and video-image weight sharing [33]. In addition, distillation-based methods improve representation learning via multi-scale teachers [1], video-image teachers [34], and asymmetric masking teachers [40]. However, the above methods may neglect dynamic masking analysis for representation learning, which could hinder the capture of complex video semantics. Therefore, it remains a key challenge to capture complex semantics in masked visual modeling.

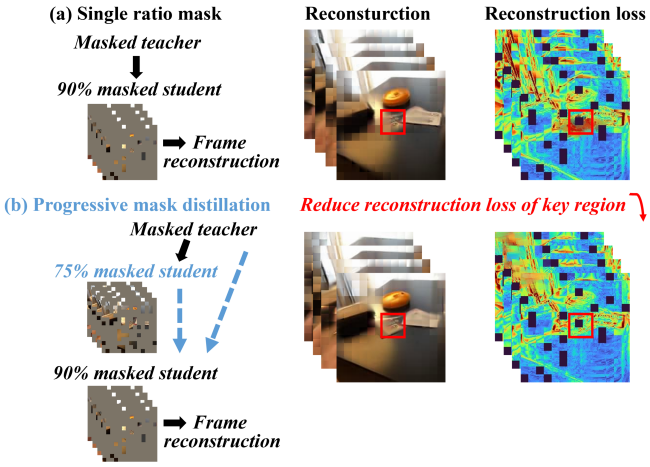


Figure 1. Illustration of progressive mask distillation for self-supervised video representation learning. (a) A single ratio mask may overlook some details in visual reconstruction. (b) Our progressive mask distillation leverages dynamic mask ratios to analyze visual features, thereby enhancing detail reconstruction.

Figure 1 illustrates the video reconstruction in masked visual modeling. When a high mask ratio (e.g., 90%) is used, the model must reconstruct from a small set of visible patches. This forces the model to infer high-level semantics from insufficient context, and enables it to provide discriminative features for downstream tasks.

In **Figure (a)**, since different semantics exist among different quantities of patches, a single mask ratio may fail

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to analyze these varying patch quantities. Because of this limitation, even with teacher signals guiding the learning of high-level semantic representations, the reconstruction error in some local region remains significant (e.g., the red box in the first line). The model may fail to capture the semantics of the red box, and cannot recognize the state of objects, which may reduce the feature discriminability for action recognition.

In **Figure (b)**, to handle varying quantities of patches, we extend the single mask ratio to multiple mask ratios. Following the principle of semantic learning from easy to hard, we capture semantics by dynamically increasing mask ratios across progressive learning phases. In the first phase, a lower mask ratio typically simplifies the feature reconstruction task. Here, the teacher primarily guides the student to capture low-level semantic information (e.g., edge information between neighboring patches). In the second phase, a higher mask ratio increases the difficulty of feature reconstruction. Beyond the teacher model, the student model trained in the first phase also guides the student in the second phase. The student in the second phase incorporates the additional influence of easy, low-level semantics on hard, high-level semantic learning. In this way, the feature reconstruction error is reduced (e.g., the red box shows a smaller reconstruction error in the second line), thereby ensuring the feature discriminability for downstream tasks.

In **Figure 2**, we propose Progressive Mask Distillation (PMD), which integrates three complementary mechanisms to address three key challenges in video representation learning: insufficient semantic learning under a single mask ratio, insufficient modeling of semantic regions, and semantic inconsistency across network layers.

First, to avoid insufficient semantic learning caused by a single mask ratio, we introduce a progressive student distillation module. Following the principle of feature reconstruction from easy to hard, this module learns semantics in multiple phases. The early-phase uses a lower mask ratio, thereby retaining more visible tokens. This facilitates low-level semantic learning (among neighboring tokens). To inherit low-level semantics, the late-phase model is initialized with the parameters of the early-phase model. After the late-phase model adopts a higher mask ratio, the original masked teacher guides it to capture key high-level semantics. Beyond this, the early-phase model also serves as an auxiliary teacher, which can leverage its learned low-level semantics to facilitate the late-phase model's high-level semantic learning. This progressive learning strategy ensures that semantic learning progresses from easy to hard.

Second, to analyze the semantic learning of key regions, we design a difficulty-aware region enhancement module. The accuracy of a region representation can be measured by its reconstruction loss. The large reconstruction losses tend to indicate the difficulty-aware regions (e.g., key regions with insufficient representation), thereby hindering their semantic learning. Since only partial regions can be covered by masks in a single epoch, it is difficult to estimate the reconstruction accuracy of more masked patches directly. To address this, the regions' reconstruction losses are memorized periodically. Within each period, one region may have different reconstruction losses across multiple epochs. The sequence of these losses is smoothed to ensure stable reconstruction estimation, which further enables the identification of difficulty-aware regions. To focus on optimizing these difficulty-aware regions, the losses are used to learn region weights. These weights can guide the teacher model to enhance the semantic learning of these key regions.

Third, to guide semantic learning across network layers, we design a cross-layer alignment module. In the base distillation network, the teacher only guides the learning of deep layers in the student network. These deep layers can reflect high-level semantic patterns. However, due to insufficient guidance for multiple layers, shallow layers may not be properly guided by this teacher. As a result, shallow layers only capture low-level semantics, and may be insufficient to support the explanation of high-level semantic patterns. Since different layers may capture different semantics, this module introduces a layer-wise cross-alignment by dividing the network into shallow, middle, and deep layers. This mechanism helps avoid large differences between shallow and deep layers. Specifically, it further guides learning in this student network by aligning features across deeper layers within this teacher network. This reduces the semantic gap between different layers, thereby enhancing semantic learning for video representations.

Our contributions are summarized as follows:

(1) We propose a Progressive Mask Distillation (PMD) model, which focuses on enhancing the self-supervised video representation via a dynamic masking strategy. To learn video representation from easy to hard, PMD designs a progressive student distillation module. It trains multiple student networks by dynamically increasing mask ratios in multiple phases . The early-phase student helps to guide the high-level semantic learning in the late-phase student.

(2) To analyze semantic learning on key regions, we design a difficulty-aware region enhancement module. It identifies difficulty-aware regions via region reconstruction loss. A difficulty-aware region loss is introduced to enhance the semantic learning of these regions.

(3) To analyze semantic consistency across multiple layers, we design a cross-layer feature alignment module. It divides the student network into shallow, middle, and deep layers to capture different levels of semantics. An alignment loss is employed to guide the shallower student layers using the deeper teacher layers, thereby enhancing the semantic learning in this student network.

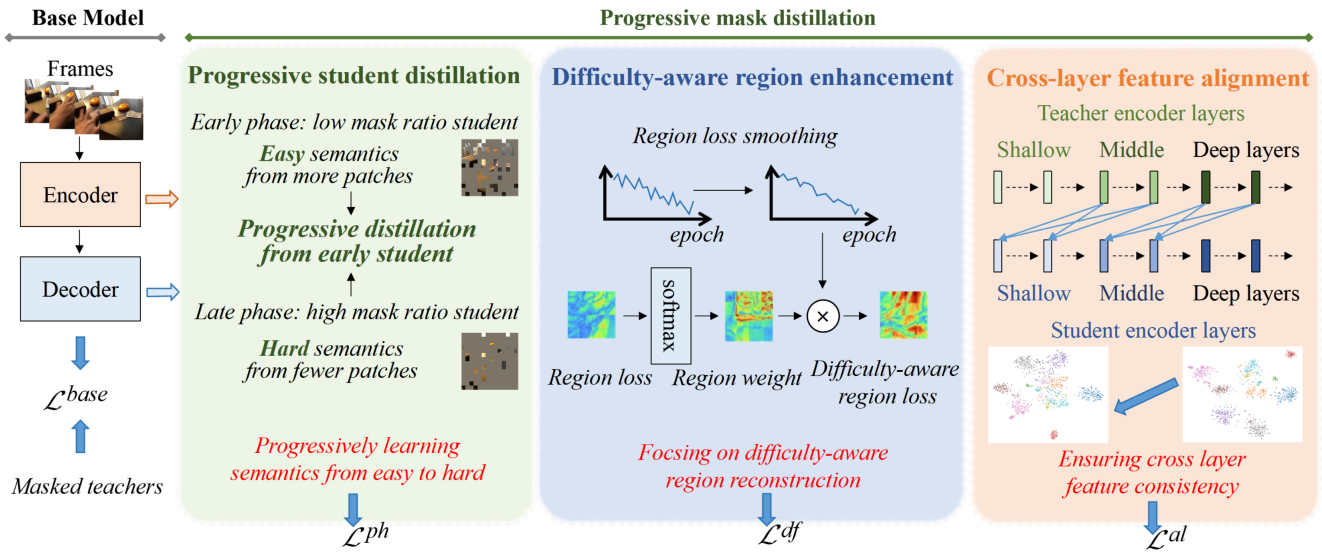


Figure 2. The overview of Progressive Mask Distillation (PMD). PMD addresses three key challenges in self-supervised video representation learning, i.e., insufficient semantic learning under a single mask ratio, insufficient modeling of semantic regions, and semantic inconsistency across network layers. These challenges are alleviated via three corresponding modules, i.e., progressive student distillation, difficulty-aware region enhancement, and cross-layer feature alignment.

2. Related Works

Masked Visual Modeling. Masked visual modeling is a self-supervised learning task that does not rely on category-specific annotations. The goal of this task is to learn discriminative visual representations by reconstructing masked patches from visible patches. For image representation learning, existing methods mainly focus on spatial masking strategies, such as tube masking [30] and running cell masking [32]. The masked learning can be enhanced via diffusion strategies [35] or cluster assignments [23]. For video representation learning, some methods adopt spatiotemporal masking [8] or redundant spatiotemporal token masking [14]. Additional modalities can enhance masked reconstruction through text-guided masking [6] and motion-aware reconstruction [37]. Beyond masking strategies, extra motion information can boost masked self-supervised learning, for instance, through dense motion trajectories [28], optical flow-guided motion [12], motion correspondence masking [5], and synthetic motion data [29]. Additionally, multi-modality fusion enhances the masked features, such as image-video weight sharing [33] and image-video decoder sharing [9]. However, the aforementioned methods tend to overlook the impact of semantic granularity (low-level vs. high-level) on representation learning, which can lead to insufficiently semantic representations. To address this, our approach guides semantic learning using dynamic mask ratios, allowing the low-level semantics learned in early-phase to guide high-level semantic learning in subsequent phase .

Distillation Learning. Distillation learning aims to transfer knowledge from a teacher model to guide the training of the student model. For general video tasks, distillation can improve various tasks, including video place recognition [25], long-tailed video recognition [18], and pseudo-caption generation for videos [39]. The distillation can be enhanced via self-distillation and cross-distillation modes [26], contrastive distributions across models [36], and hierarchical knowledge transfer [21]. For masked video modeling, some methods consider single-teacher knowledge distillation, such as unmasked teacher distillation [17] and intra-class knowledge distillation [19]. Self-distillation can be enhanced by incorporating motion-aware weights [22] and temporal masked features [38]. Distillation can also be guided by fusing features from multiple teachers, such as different-scale masked teachers [1], asymmetric masked teachers [40], and image-video masked teachers [34]. Knowledge distillation can further enhance contrastive language-image pretraining [3, 24]. Multi-stage learning can improve distillations, for example, distillation from a pre-trained masked teacher and a fine-tuned teacher [13], or using an ensemble of multiple students [2]. Cross-layer analysis helps enhance multi-stage distillation, such as cross-layer knowledge distillation [7], distillation from the same teacher model [41], and distillation with deeper student models [4]. Unlike the above methods, our model aims to enable semantic learning from easy to hard by dynamically increasing mask ratios in progressive student learning, thereby ensuring the semantic-aware video representation.

3. Method

Figure 2 shows the overview of the Progressive Mask Distillation (PMD) model. PMD is built on a Transformer-based model, which originally learns semantics only under a high mask ratio. This may lead to three key challenges, including insufficient semantic learning under a single mask ratio, insufficient modeling of semantically regions, and semantic inconsistency across network layers. To address these issues, PMD introduces three corresponding modules, i.e., the progressive student distillation module, the difficulty-aware region enhancement module, and the cross-layer feature alignment module.

3.1. Base Model

The base model adopts masked video distillation [34] for self-supervised video representation learning. It optimizes a masked video student using two masked teachers: an image teacher [11] and a video teacher [30]. For each teacher, the corresponding distillation loss encourages the student's features to align with the teacher's features. Given the image decoder features $X^{\text{img}} = \{x_p^{\text{img}}\}$ and the student decoder features $X^{\text{st}} = \{x_p^{\text{st}}\}$ (where p denotes the index of masked patches), the image teacher loss employs the L2 loss, defined as $\mathcal{L}_{\text{base}}^{\text{img}} = \frac{1}{N_M}\sum_{p\in M} \|x_p^{\text{img}} - x_p^{\text{st}}\|_2^2$. Here, M is the set of masked tokens, and N_M is the number of masked tokens. Similarly, given the video decoder features $X^{\text{vid}} = \{x_p^{\text{vid}}\}$, the video teacher loss is $\mathcal{L}_{\text{base}}^{\text{vid}} = \frac{1}{N_M}\sum_{p\in M} \|x_p^{\text{vid}} - x_p^{\text{st}}\|_2^2$. The total base loss is thus:$\mathcal{L}^{\text{base}} = \mathcal{L}_{\text{base}}^{\text{img}} + \mathcal{L}_{\text{base}}^{\text{vid}}$.

Discussion. The base model performs distillation for a student trained with a single mask ratio, but a single mask ratio may be insufficient to reconstruct detailed features. In this work, we aim to enhance feature learning by progressively increasing mask ratios across multiple students. The early-phase student uses a low mask ratio, and reconstructs features with many visual tokens to learns low-level semantics. When the early-phase student guides the late-phase student (which uses a high mask ratio), the late-phase student leverages the learned low-level semantics to preserve detailed features while capturing high-level semantics.

3.2. Progressive Student Distillation

The teacher in the base model analyzes the reconstruction loss of masked patches with a high mask ratio. When the high mask ratio masks some key patches, this teacher may fail to provide an accurate video representation. In particular, some complex semantics are related to multiple contexts from more visible patches. Thus, we design Progressive Student Distillation module to capture different semantics by dynamically increasing mask ratios in multi-phase student learning. The early-phase student uses a low mask ratio to retain more visible patches, leveraging these patches to learn easy, low-level semantics. The late-phase student uses a high mask ratio to eliminate redundant visible patches, relying on limited visible patches to learn hard, high-level semantics. To ensure semantic learning from easy to hard, the late-phase student is initialized with the pre-trained parameters of the early-phase student. Moreover, beyond distilling knowledge from the base teacher model, the late-phase student is also guided by the early-phase student.

In the first phase, the student only uses two teachers. Since no additional teacher is available from previous phases (as Phase 1 is the initial stage), we set the progressive loss to its default value of 0. In subsequent phases (i.e., $ph > 1$), the student model trained in Phase $ph - 1$ serves as an additional teacher to guide feature learning for the current phase. Using the student decoder features of Phase $ph - 1$ ($X^{ph-1} = \{x_p^{ph-1}\}$), the progressive loss trains the student decoder features of Phase ph ($X^{st} = \{x_p^{st}\}$) at masked position p as:

$$\mathcal{L}^{pg} = \begin{cases} 0 & if\ ph = 1 \\ \frac{1}{N_M}\sum_{p\in M} ||x_p^{ph-1} - x_p^{st}||_2^2 & if\ ph > 1 \end{cases} \quad (1)$$

3.3. Difficulty-Aware Region Enhancement

The Difficulty-Aware Region Enhancement further analyzes the semantic learning of key patches. Key patches usually have complex semantics. When these key patches have large patch losses, their semantic information may not be captured accurately. To ensure effective semantic learning for difficult patches, we estimate region-level weights based on patch losses. The region-level weights are updated periodically (i.e., grouped every several epochs within a stage), which gradually identifies difficulty-aware regions.

In masked visual modeling, masks are generated randomly at different positions across epochs. In the student network, patch losses are not calculated for visible patches and are initialized to 0. For masked patches, the patch loss is initialized as the L2 loss between the student decoder features and the teacher decoder features. In the image teacher branch, the patch loss at epoch i is defined as $\mathcal{L}_{i,p}^{img} = ||x_p^{img} - x_{i,p}^{st}||_2^2$. To avoid large fluctuations, the patch loss is updated via exponential moving average to smooth out noise in loss estimation. To update the patch loss for masked patch p at the current epoch i, we first locate the most earlier epoch i' where patch p was also masked. The smoothed patch loss is then calculated as $\hat{\mathcal{L}}_{i,p}^{img} = \alpha \cdot \mathcal{L}_{i,p}^{img} + (1-\alpha) \cdot \hat{\mathcal{L}}_{i',p}^{img}$, where α denotes the update weight.

To learn the patch weights dynamically, we first group N_h epochs into each stage. In the image teacher branch, the patch weights $w_{1,p}^{img} = 1/N_M$ are initialized to the same in the first stage $s = 1$. For stages $s > 1$, we denote the smoothed patch loss at the final epoch of the previous stage $(s-1)$ as $\hat{\mathcal{L}}_{s-1,p}^{img}$. This stage-wise loss is

then smoothed using the softmax function to learn the patch weights as $w_{s,p}^{img} = \frac{\exp(\hat{\mathcal{L}}_{s-1,p}^{img}/\tau)}{\sum_k \exp(\hat{\mathcal{L}}_{s-1,k}^{img}/\tau)}$. Here, τ denotes a temperature parameter that helps stabilize the weight learning process. The learned weights are used to reweight the patch losses, and the reweighted patch losses are summed to form the difficulty-aware region loss. The total difficulty-aware region loss incorporates two teachers, defined as $\mathcal{L}^{df} = \mathcal{L}_{df}^{\text{img}} + \mathcal{L}_{df}^{\text{vid}}$. Each loss is calculated similarly as:

$$\begin{cases} \mathcal{L}_{df}^{\text{img}} = \frac{1}{N_M} \sum_{p \in M} w_{s,p}^{img} \hat{\mathcal{L}}_{i,p}^{img} \\ \mathcal{L}_{df}^{\text{vid}} = \frac{1}{N_M} \sum_{p \in M} w_{s,p}^{vid} \hat{\mathcal{L}}_{i,p}^{vid} \end{cases} \tag{2}$$

3.4. Cross-Layer Feature Alignment

The Cross-Layer Feature Alignment module enhances features through dense distillation. Following [34], we divide the Transformer encoder layers to characterize features with different semantic information. Shallow layers typically describe video patterns that align with original patch details (e.g., color, edge, texture). Middle layers usually characterize patterns by balancing detail and semantics (e.g., object parts, entire objects). Deep layers generally capture high-level semantic patterns (e.g., object interactions, action or scene semantics). Beyond basic knowledge distillation, this shallow-middle-deep alignment can avoid the large semantic gap that would arise from direct alignment between shallow and deep layers.

For consistent feature distillation, we first set the shallow/middle/deep layer sets $\mathbb{H}, \mathbb{M}, \mathbb{D}$. The student features are aligned with the teacher features for each masked patch p. Given the student feature at encoder layer n_s, we use a fully connected (FC) layer (with non-shared parameters across layers) to learn an student feature $\hat{z}_p^{n_s} = \text{FC}(z_p^{n_s})$, which has the same dimension of the teacher feature $z_p^{n_t}$. For a masked patch p, we ensure consistency between the student's shallow layers and the teacher's middle layers via the layer-pair difference $d_p^{\mathbb{H}\to\mathbb{M}} = \frac{1}{N_{\mathbb{H}} N_{\mathbb{M}}} \sum_{n_s \in \mathbb{H}} \sum_{n_t \in \mathbb{M}} \|\hat{z}_p^{n_s} - z_p^{n_t}\|_2^2$. Similarly, to ensure consistency between the student's middle layers and the teacher's deep layers, their layer-pair difference is $d_p^{\mathbb{M}\to\mathbb{D}} = \frac{1}{N_{\mathbb{M}} N_{\mathbb{D}}} \sum_{n_s \in \mathbb{M}} \sum_{n_t \in \mathbb{D}} \|\hat{z}_p^{n_s} - z_p^{n_t}\|_2^2$. $N_{\mathbb{H}}, N_{\mathbb{M}}, N_{\mathbb{D}}$ denote the numbers of $\mathbb{H}/\mathbb{M}/\mathbb{D}$ layer sets, respectively. Both the image and video teachers follow the same consistency learning process. The total cross-layer feature alignment loss considers two teachers as $\mathcal{L}^{al} = \mathcal{L}_{\text{al}}^{\text{img}} + \mathcal{L}_{\text{al}}^{\text{vid}}$. Each loss is calculated as:

$$\begin{cases} \mathcal{L}_{\text{al}}^{\text{img}} = \frac{1}{N_M} \sum_{p \in M} (d_p^{\mathbb{H}\to\mathbb{M},img} + d_p^{\mathbb{M}\to\mathbb{D},img}) \\ \mathcal{L}_{\text{al}}^{\text{vid}} = \frac{1}{N_M} \sum_{p \in M} (d_p^{\mathbb{H}\to\mathbb{M},vid} + d_p^{\mathbb{M}\to\mathbb{D},vid}) \end{cases} \tag{3}$$

3.5. Model Optimization

Following [34], during pre-training, our model is trained in a self-supervised manner that does not rely on category labels. In phase ph, we group several epochs into multiple stages s to learn smoothed patch losses. The total loss is defined as $\mathcal{L}^{total} = \lambda^{\text{base}}\mathcal{L}^{\text{base}} + \lambda^{pg}\mathcal{L}^{pg} + \lambda^{df}\mathcal{L}^{df} + \lambda^{al}\mathcal{L}^{al}$, where λ^{base}, λ^{pg}, λ^{df}, and λ^{al} denote the hyper-parameters that balance the weights of each term. The final output model is the student network from the last phase. **Algorithm 1** shows the PMD pre-training algorithm. For fine-tuning, we append a single-layer FC classifier to our model. The model is fully fine-tuned via cross-entropy loss.

Algorithm 1: PMD pre-training algorithm

Require: Training frames, teacher models, student mask ratios, phase number, epoch number in each stage.
Ensure: Trained student model in multiple phase.

```
Learn teacher's encoded/decoded features.
for each training phase do
    Initialize student model in phase ph.
    for each training stage do
        Learn student's encoded/decoded features.
        Learn base loss L^base
        Learn progressive loss L^pg
        Learn difficulty-aware region loss L^df
        Learn alignment loss L^al.
        Train student model in phase ph with L^total.
    end for
    Output the trained student models in phase ph.
end for
```

4. Experiments

4.1. Datasets and Implementation Details

Datasets. We conduct experiments on four video datasets. **Something-Something V2 (SSv2)** [10] contains 220,847 video clips covering 174 action categories, which focus on fine-grained human-object interaction scenarios. **Kinetics-400 (K400)** [15] contains approximately 300k video clips spanning 400 action categories. These categories include human actions, human-human interactions, and human-object interactions. **UCF101** [27] consists of 13,320 video clips across 101 action categories. **HMDB51** [16] comprises 6,766 video clips covering 51 action categories. For above datasets, we adopt the training/testing splits as specified in [29, 30].

Metrics. Following [30], we report Top-1 and Top-5 accuracy for all datasets. The FLOPs are reported to measure the computational cost of model complexity.

Implementation details. Following [34], our PMD employs two distillation teachers: the image teacher pre-trained on ImageNet-1K dataset, and the video teacher pre-trained on K400 dataset. Our student model is pre-trained on K400 dataset. Our model can be applied to different encoders and decoders. Following [34], when the stu-

Table 1. Comparison with state-of-the-art self-supervised video representation method on Something-Something V2. GFLOPs are reported for multiple clip × multiple crop in model testing.

Method	Backbone	GFLOPs	Param	Top-1(%)
VideoMAEV1$_{2400e}$ [30]	ViT-S	$57 \times 2 \times 3$	22M	66.8
AMD$_{800e}$ [40]	ViT-S	$57 \times 2 \times 3$	22M	70.2
PMD$_{400e}$ (ours)	ViT-S	$57 \times 2 \times 3$	22M	**70.7**
VideoMAEV1$_{800e}$ [30]	ViT-B	$180 \times 2 \times 3$	87M	69.6
DMAE$_{800e}$ [1]	ViT-B	$180 \times 2 \times 3$	87M	70.0
MME [28]	ViT-B	$180 \times 2 \times 3$	87M	70.5
BEVT[33]	Swin-B	$321 \times 1 \times 3$	88M	70.6
VideoMAEV1$_{2400e}$ [30]	ViT-B	$180 \times 2 \times 3$	87M	70.8
VideoMAEV2$_{1200e}$ [32]	ViT-B	$180 \times 2 \times 3$	87M	71.2
MGM [5]	ViT-B	$180 \times 2 \times 3$	87M	70.6
SIGMA [23]	ViT-B	$180 \times 5 \times 3$	87M	71.2
MGMAE$_{2400e}$ [12]	ViT-B	$180 \times 2 \times 3$	87M	72.3
SMILE$_{800e}$[29]	ViT-B	$200 \times 2 \times 3$	94M	72.5
MVD$_{400e}$ [34]	ViT-B	$180 \times 2 \times 3$	87M	72.5
AMD$_{800e}$ [40]	ViT-B	$180 \times 2 \times 3$	87M	73.3
PMD$_{400e}$ (ours)	ViT-B	$180 \times 2 \times 3$	87M	**73.7**
VideoMAEV1$_{2400e}$ [30]	ViT-L	$597 \times 2 \times 3$	305M	74.3
VideoMAEV2$_{1200e}$ [32]	ViT-L	$597 \times 2 \times 3$	305M	75.7
MVD$_{400e}$ [34]	ViT-L	$597 \times 2 \times 3$	305M	76.1
PMD$_{400e}$ (ours)	ViT-L	$597 \times 2 \times 3$	305M	**76.4**

dent model uses a ViT-S/B/L encoder, the decoder adopts a 2-layer/2-layer/4-layer cross-attention Transformer. Our model takes 16 frames as input (for ViT-S/B/L). Each frame is resized to have a shorter side of 256 pixels [30], then cropped to 224 × 224. Cube masking [30] is used to generate masked video tokens. In progressive student distillation for pre-training, we use four phases (each 100 epochs) with mask ratios increasing from low to high ($75\% \rightarrow 80\% \rightarrow 85\% \rightarrow 90\%$). In difficulty-aware region enhancement, we divide every 32 epochs into one stage, with a weight update $\alpha = 0.95$ and a temperature parameter $\tau = 0.5$. For cross-layer feature alignment, since ViT-S and ViT-B both have 12 layers, we define the shallow layers as layers 1–4, middle layers as layers 5–8, and deep layers as layers 9–12. For ViT-L (24 layers), we set the shallow layers as layers 1–8, middle layers as layers 9–16, and deep layers as layers 17–24. The total loss uses $\lambda^{\text{base}} = 1$, $\lambda^{pg} = 0.05$, $\lambda^{df} = 1$, and $\lambda^{al} = 1$. Following [34], the training epochs for full fine-tuning are set as 150/75/50 for ViT-S/B/L on K400, 40/30/30 for ViT-S/B/L on SSv2, 100 for ViT-B on UCF101, and 50 for ViT-B on HMDB51. AdamW [20] uses an initial learning rate of 1e-3 for ViT-S/B/L on K400, 1e-3/5e-4/5e-4 for ViT-S/B/L on SSv2, 5e-4 for ViT-B on UCF101, and 1e-3 for ViT-B on HMDB51. The batch size is 16/8/4 for ViT-S/B/L.

4.2. Comparison with State-of-the-Art

Results on SSv2 and K400. Table 1 and Table 2 show the comparisons on SSv2 and K400 datasets, respectively. Some methods adopt different masking strategies, i.e., tube masking in VideoMAE V1 [30], and running cell masking in VideoMAE V2 [32]. SMILE [29] fuses spatial and motion semantics by adding synthetic motion training data. Other methods leverage knowledge distillation, such as image-video masked teacher models in MVD [34], and asymmetric masked distillation in AMD [40]. Unlike the above methods, our PMD progressively increases the mask ratio to train multiple student models. It introduces a progressive distillation strategy to perform semantic distillation from easy to hard, thereby boosting the performance.

Table 2. Comparison with state-of-the-art self-supervised video representation method on Kinetics-400. GFLOPs are reported for multiple clip × multiple crop in model testing.

Method	Backbone	GFLOPs	Param	Top-1(%)
VideoMAEV1$_{1600e}$[30]	ViT-S	$57 \times 5 \times 3$	22M	79.0
AMD$_{800e}$[40]	ViT-S	$57 \times 5 \times 3$	22M	80.9
PMD$_{400e}$ (ours)	ViT-S	$57 \times 5 \times 3$	22M	**81.2**
VideoMAEV1$_{800e}$ [30]	ViT-B	$180 \times 5 \times 3$	87M	80.0
BEVT[33]	Swin-B	$282 \times 4 \times 3$	88M	80.6
OmniMAE [9]	ViT-B	$180 \times 5 \times 3$	87M	80.8
DMAE$_{800e}$ [1]	ViT-B	$180 \times 5 \times 3$	87M	80.8
MAEST[8]	ViT-B	$180 \times 7 \times 3$	87M	81.3
VideoMAEV1$_{1600e}$ [30]	ViT-B	$180 \times 5 \times 3$	87M	81.5
VideoMAEV2$_{1200e}$ [32]	ViT-B	$180 \times 5 \times 3$	87M	81.5
MME [28]	ViT-B	$180 \times 7 \times 3$	87M	81.8
MGM [5]	ViT-B	$180 \times 2 \times 3$	87M	80.8
SIGMA [23]	ViT-B	$180 \times 5 \times 3$	87M	81.5
MGMAE$_{1600e}$ [12]	ViT-B	$180 \times 5 \times 3$	87M	81.8
AMD$_{800e}$ [40]	ViT-B	$180 \times 5 \times 3$	87M	82.2
MVD$_{400e}$ [34]	ViT-B	$180 \times 5 \times 3$	87M	82.7
SMILE$_{800e}$[29]	ViT-B	$200 \times 7 \times 3$	94M	83.1
PMD$_{400e}$ (ours)	ViT-B	$180 \times 5 \times 3$	87M	**83.3**
DiffMAE [35]	ViT-L	$598 \times 7 \times 3$	304M	84.5
VideoMAEV1$_{1600e}$ [30]	ViT-L	$597 \times 5 \times 3$	305M	85.2
VideoMAEV2$_{1200e}$ [32]	ViT-L	$597 \times 5 \times 3$	305M	85.4
MVD$_{400e}$ [34]	ViT-L	$597 \times 5 \times 3$	305M	86.0
PMD$_{400e}$ (ours)	ViT-L	$597 \times 5 \times 3$	305M	**86.3**

Table 3. The effect of feature transferability with Top-1(%). The model without Teacher only uses K400 for pre-training.

Method	K400	SSv2	UCF101	HMDB51
without Teacher				
VideoMAEv1 [30]	80.0	69.6	91.3	62.6
VideoMAEv2 [32]	81.5	71.2	92.5	63.8
SMILE [29]	83.1	72.1	96.4	69.5
with Teacher				
MVD (IN1K+K400) [34]	82.7	72.5	97.0	76.4
Ours (IN1K)	83.1	73.2	97.5	78.2
Ours (K400)	82.9	73.5	97.3	78.4
Ours (IN1K+K400)	**83.3**	**73.7**	**97.6**	**78.6**

The effect of feature transferbility for UCF101 and HMDB51. Table 3 shows the effect of feature transferability. First, our PMD model is pre-trained on the K400 dataset to learn discriminative features. Then, the pre-trained model is fully fine-tuned on different downstream datasets. Equipped with dual teachers (image + video), our model outperforms MVD on all downstream datasets (i.e., K400, SSv2, UCF101, HMDB51). This indicates that the learned semantic features possess strong transferability across diverse downstream datasets.

4.3. Ablation study

We report the ablation study on SSv2 with a ViT-B backbone pretrained on K400.

Table 4. The effect of PMD. Training times of self-supervised model are reported by splitting 400 epochs into multiple phases.

Mask ratios in phases	Epochs	Top-1 (%)
75%	400	72.7
80%	400	72.8
85%	400	72.9
90%	400	73.1
75% → 90%	200+200	73.4
75% → 80% → 90%	100+100+200	73.6
75% → 85% → 90%	100+100+200	73.5
75% → 80% → 85% → 90%	100+100+100+100	**73.7**

Table 5. The effect of difficulty-aware region enhancement. SPL, RW denote smoothed patch loss, and region weight.

	without	SPL	RW	SPL+ RW
Top-1(%)	72.9	73.2	73.4	**73.7**

Table 6. The effect of epoch number in stage.

Epoch number in stage	8	16	32	64
Top-1 (%)	73.4	73.6	**73.7**	**73.7**

Table 7. The effect of cross layer feature alignment. $\mathbb{HM}$, $\mathbb{MD}$ denote Shallow → Middle, and Middle → Deep alignment.

	without	$\mathbb{HM}$	$\mathbb{MD}$	$\mathbb{HM}$+$\mathbb{MD}$
Top-1(%)	73.2	73.4	73.3	**73.7**

The effect of PMD. Table 4 presents the performance of different progressive masking student models. The training time of self-supervised model is reported over 400 epochs on 4 NVIDIA RTX 3090 GPUs. The single-phase model adopts a fixed mask ratio. The model with a high mask ratio has a shorter training time compared with that with a low mask ratio. A high mask ratio (e.g., 90%) forces the model to infer high-level semantics from limited patches, achieving a Top-1 accuracy of 73.1%. In contrast, the multi-phases model divides the training process into multiple phases. For the two-phase setting (with the mask ratio increasing from 75% to 90%), the early-phase (75%) enables the model to learn low-level semantics from abundant visible patches. Subsequent distillation (guided by these learned low-level semantics) further enhances high-level semantic learning in the late-phase (90%). Comparing the two three-phase settings (75% → 80% → 90% vs. 75% → 85% → 90%), the former enables more robust low-level semantic learning by employing a smoother progression in the early phase. To further smooth the transition from low-level to high-level semantics, the four-phase distillation (75% → 80% → 85% → 90%) achieves the best performance.

The effect of difficulty-aware region enhancement. In **Table 5**, we use smoothed patch loss (SPL) to reduce the noise in patch-wise loss estimation. We then learn region weights (RW) from the SPL, which enables the model to focus on difficulty-aware regions. By jointly applying these two strategies, the model achieves the best performance.

Table 8. The effect of model modules, i.e., Base, Progressive Student Distillation (PSD), Difficulty-Aware Region Enhancement (DARE), and Cross-Layer Feature Alignment (CLFA).

Base	PSD	DARE	CLFA	Top-1(%)
✓				72.5
✓	✓			73.0
✓		✓		72.8
✓			✓	72.7
✓	✓	✓		73.5
✓		✓	✓	73.1
✓	✓	✓	✓	**73.7**

Table 9. Params, FLOPs, GPU hour, and Wall-Clock Time (WCT).

Models	Top-1(%)↑	Params(M)↓	FLOPs(G)↓	GPU(H)↓	WCT(H)↓
MVD	72.5	120.56	116480	1284	321
+PSD	73.0	120.56	132160	1410	352
+DARE	72.8	120.56	116480	1283	321
+CLFA	72.7	130.00	117520	1314	328
PMD	73.7	130.00	133200	1440	360
VideoMAEV1	69.1	94.2	44560	408	102

Table 10. Effect of pre-training epoch.

Pre-training epoch	100	200	400	800
Top-1(%)	71.4	72.2	**73.7**	**73.7**

Table 11. The effect of teacher models on ViT-S student.

ViT-B Teacher	Image	Video	Image+Video
Top-1(%)	73.1	73.4	**73.7**

The effect of epoch number in stage. To stabilize the calculation of the region loss, the smoothed patch loss is updated by grouping multiple epochs into a single stage. In **Table 6**, more epochs per stage contribute to a smoother region loss, and the model converges when the number of epochs per stage reaches 32.

The effect of cross layer feature alignment. In **Table 7**, the student's shallow layers are guided by the teacher's middle features, while the student's middle layers are guided by the teacher's deep features. By jointly applying these two-step guidelines, the model achieves the best performance.

The effect of model modules. **Table 8** shows the effect of four modules, i.e., Base, Progressive Student Distillation (PSD), Difficulty-Aware Region Enhancement (DARE), and Cross-Layer Feature Alignment (CLFA). **(1)** The base model uses the base loss $\mathcal{L}^{Base}$, which guides the student's feature learning to approximate the image and video teacher features. **(2)** To enable "easy-to-hard" semantic learning, the PSD module initializes the late-phase student with the pre-trained parameters of the early-phase student. The early-phase student uses a low mask ratio to learn low-level semantics among multiple patches, while the late-phase student uses a high mask ratio to learn high-level semantics from limited patches. Correspondingly, the progressive loss $\mathcal{L}^{pg}$ leverages the early-phase student (as an additional teacher) to guide the late-phase student's feature

learning, fusing details from different mask ratios. **(3)** The DARE module uses the difficulty-aware region loss $\mathcal{L}^{df}$ to focus on the accurate semantics of key regions. This addresses the issue of insufficient learning for high-loss difficult regions. **(4)** The CLFA module uses the alignment loss $\mathcal{L}^{al}$ to guide shallow feature learning. It bridges the semantic gap between shallow, middle, and deep features. By jointly incorporating the above modules, our model achieves the best performance. As well, in Table 9, we integrating "PSD, DARE, CLFA" modules brings a 1.2% Top-1 accuracy gain with only 114% FLOPs overhead. This gain verifies our systematically synergistic design rather than isolated techniques.

The effect of pre-training epochs. Table 10 shows the effect of pre-training epochs. The total pre-training epochs are equally divided into four phases (corresponding to four students). More epochs enable the model to capture detailed patch features. The accurate pre-trained features further reduce the difficulty of fine-tuning, and improve the performance on downstream tasks (e.g., Top-1 accuracy on SSv2). Notably, the model performance converges when the pre-training epochs reach 800 (200 epochs per phase).

The effect of teacher modality. Table 11 shows the effect of different teachers on the student. For the ViT-S teachers, the dual-modality teachers (image and video) provide rich complementary features for semantic learning, outperforming single-modality teachers (either image-only or video-only).

5. Visualization

Visualization of feature distribution with t-SNE. Figure 3 visualizes the 2-dimensional feature space of SSv2 categories using t-SNE [31]. In (a), the base model [34] enhances the masked student with image-video teachers and fine-tunes the masked features for action recognition. In (b), the progressive model uses image-video teachers and learns semantic features from multiple students via multi-phase learning. Early-phase students with lower mask ratios learn low-level semantics, which guide subsequent students to distill high-level semantics. This progressive learning ensures effective feature separation. In (c), the progressive region model additionally uses difficulty-aware region enhancement to identify key regions. By strengthening semantic learning in these key regions, feature separation is further improved. In (d), the full model enhances semantic learning with additional cross-layer feature alignment. The student features are divided into shallow, middle, and deep layers to capture semantics at different levels. Specifically, the shallow student features are guided by the middle teacher features, while the middle student features are guided by the deep teacher features. Integrating all the above strategies, the full model achieves the best feature separation.

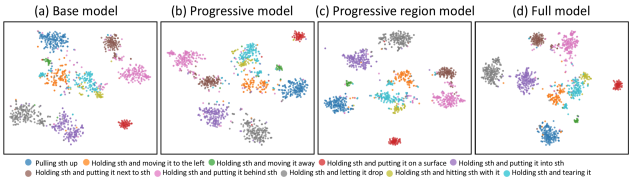


Figure 3. Feature distribution visualization of PMD model.

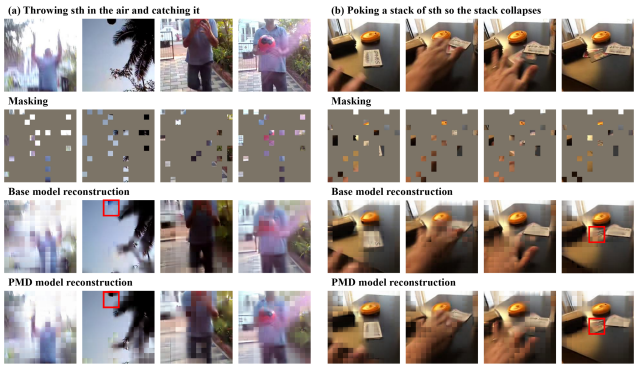


Figure 4. Frame reconstruction of PMD model.

Visulization of frame reconstruction. Figure 4 visualizes two instances to analyze the reconstruction of key regions for action recognition. In (a), the object being thrown is a critical region. The base model adopts a high mask ratio, which may overlook fine-grained details (see the red box in Column 2). Our model uses an early-phase student with a lower mask ratio (retaining more neighboring visible patches) to learn low-level semantics. Then, a late-phase student with a high mask ratio (retaining limited patches) focuses on high-level semantics. To ensure the accurate reconstruction of key regions, the late-phase student is guided by both the original teacher and the early-phase student. In (b), the object collapsing due to hand poking is another critical region. By following the same progressive student distillation logic, our model ensures the reconstruction of this key region (see the red box in Column 4), further enabling accurate semantic capture to support action recognition.

6. Conclusion

This work proposes a Progressive Mask Distillation (PMD) model to address the limitations of a fixed single mask ratio in masked self-supervised video representation learning. We design a progressive student distillation strategy, which uses dynamic mask ratios across multiple phases to enable semantic learning from easy to hard. Additionally, we design a difficulty-aware region enhancement module to reinforce the semantic learning of key regions. We design cross-layer feature alignment to ensure consistent feature learning across different layers. By integrating the above three components, our model outperforms state-of-the-art methods.

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